



MANUAL PEMELIHARAAN SISTEM (MAINTENANCE MANUAL)

FACILITY MANAGEMENT SERVICE PLATFORM (FMSP) LINTASARTA

Dokumentasi Teknis, Konfigurasi, Deployment, Backup & Restore
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PT Applikanusa Lintasarta
Divisi IT Infrastructure & DevOps

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1. Pendahuluan & Lingkup Sistem

Manual Pemeliharaan ini ditujukan untuk **administrator sistem, tim IT Infrastructure, dan DevOps** PT Aplikanusa Lintasarta. Dokumen ini berisi instruksi teknis lengkap untuk mengelola, mengonfigurasi, melakukan pencadangan data, migrasi skema database, dan memelihara infrastruktur aplikasi FMSP Lintasarta agar berjalan optimal.

 **RAHASIA:** Dokumen ini mengandung informasi sensitif tentang arsitektur keamanan, kredensial default, dan konfigurasi infrastruktur. Dilarang menyebarluaskan tanpa persetujuan Divisi Keamanan Informasi.

2. Arsitektur & Tech Stack

2.1 Komponen Utama

Layer	Teknologi	Versi
Frontend Framework	Next.js (App Router)	16.2.9
UI Library	React	19.2.4
Language	TypeScript	5.x
CSS Framework	Tailwind CSS	4.x
ORM	Prisma Client	6.19.3
Database	PostgreSQL	16+
Auth — JWT	jsonwebtoken	9.0.3
Auth — SSO	Google OAuth API	—
Auth — Hashing	bcryptjs	3.0.3
Email	Nodemailer	9.0.0
Cron Scheduler	node-cron	4.2.1
Charts	Recharts	3.8.1
Push Notification	web-push (VAPID)	3.6.7
QR Code	qrcode	1.5.4
Validation	Zod	4.4.3
PDF Generation	@react-pdf/renderer	4.5.1
AI Copilot	Ollama (Qwen 2.5)	—

Layer	Teknologi	Versi
Realtime	Socket.IO	4.8.3
Icons	Lucide React	1.20.0
Date Utility	date-fns	4.4.0
Deployment	Fly.io + Docker	—

2.2 Struktur File Krusial

File / Direktori	Fungsi
<code>prisma/schema.prisma</code>	Definisi 22 tabel database PostgreSQL + relasi
<code>prisma/seed.ts</code>	Script seeding data awal (SuperAdmin, Master Data, AppSettings)
<code>src/lib/db.ts</code>	Singleton Prisma Client (mencegah connection pool exhaustion)
<code>src/lib/auth-middleware.ts</code>	JWT verification + brute force protection (lockout 15 min / 5 attempts)
<code>src/lib/rbac.ts</code>	Role definitions & permission matrix (6 roles)
<code>src/lib/rbac-middleware.ts</code>	Middleware enforcer per-endpoint RBAC check
<code>src/lib/validators.ts</code>	Zod schemas untuk semua input API (XSS protection)
<code>src/lib/file-validator.ts</code>	File upload validation (type whitelist, size limit 10MB, magic bytes check)
<code>src/lib/email-service.ts</code>	Nodemailer SMTP wrapper + Ethereum mock mode
<code>src/lib/push-service.ts</code>	Web Push VAPID notification sender
<code>src/lib/cron-auth.ts</code>	CRON_SECRET header verification untuk endpoint <code>/api/cron/*</code>

File / Direktori	Fungsi
<code>src/lib/region-filter.ts</code>	Prisma WHERE clause generator untuk filter region per role
<code>src/lib/api-error.ts</code>	Centralized error handler (sanitized error responses)
<code>src/app/api/</code>	11 route groups: auth, assets, legal-documents, management (11 sub-routes), analytics, audit-logs, chat, cron, notifications, reports, upload
<code>fly.toml</code>	Fly.io deployment config (region: sin, port: 8080)
<code>Dockerfile</code>	Multi-stage Docker build (node:22-alpine)
<code>docker-compose.yml</code>	Local dev setup: PostgreSQL container + app

2.3 Middleware & Library Keamanan

Setiap API endpoint dilindungi oleh chain middleware berikut:

1. **Rate Limiter:** Membatasi jumlah request per IP per window (login: 5 req/min, API: 60 req/min)
2. **JWT Verifier (auth-middleware):** Memvalidasi token dari header Authorization, cek expired, cek user aktif
3. **RBAC Enforcer (rbac-middleware):** Memeriksa apakah role user memiliki permission untuk resource + action
4. **Input Validator (validators/Zod):** Memvalidasi dan sanitize body request — strip HTML tags, escape special chars
5. **Audit Logger:** Mencatat operasi ke tabel AuditLog

3. Konfigurasi Sistem

3.1 Variabel Lingkungan (.env)

Salin file `.env.example` menjadi `.env` dan ubah semua nilai default:

Variabel	Deskripsi	Contoh / Default
<code>DATABASE_URL</code>	URL koneksi PostgreSQL	postgresql://user:pass@host:5432/db
<code>POSTGRES_USER</code>	Username PostgreSQL (Docker)	fmsp_user
<code>POSTGRES_PASSWORD</code>	Password PostgreSQL (Docker)	<i>Harus diganti!</i>
<code>POSTGRES_DB</code>	Nama database	fmsp_db
<code>JWT_SECRET</code>	Secret key penandatanganan JWT	<i>Generate: openssl rand -hex 64</i>
<code>GOOGLE_CLIENT_ID</code>	OAuth Client ID dari Google Cloud Console	xxx.apps.googleusercontent.com
<code>PORT</code>	Port aplikasi	3847 (dev) / 8080 (prod)
<code>APP_BASE_URL</code>	URL root aplikasi	https://fmsp-lintasarta.fly.dev
<code>SMTP_HOST</code>	Server SMTP	smtp.office365.com
<code>SMTP_PORT</code>	Port SMTP	587
<code>SMTP_USER</code>	Email pengirim	noreply@lintasarta.co.id

Variabel	Deskripsi	Contoh / Default
<code>SMTP_PASSWORD</code>	Password SMTP	<i>Harus diset!</i>
<code>CRON_SECRET</code>	Secret untuk autentikasi endpoint cron	<i>Generate: openssl rand -hex 32</i>
<code>NEXT_PUBLIC_VAPID_PUBLIC_KEY</code>	VAPID public key untuk Web Push	<i>Generate: npx web-push generate-vapid-keys</i>
<code>VAPID_PRIVATE_KEY</code>	VAPID private key	<i>Pair dengan public key</i>
<code>VAPID_CONTACT</code>	Email kontak VAPID	mailto:admin@lintasarta.co.id
<code>OLLAMA_BASE_URL</code>	URL server Ollama lokal	http://127.0.0.1:11434
<code>OLLAMA_MODEL</code>	Model AI yang digunakan	qwen2.5:1.5b

 **KRITIS:** Variabel `JWT_SECRET` , `CRON_SECRET` , dan `POSTGRES_PASSWORD` HARUS diganti dari nilai default. Aplikasi menolak start jika `JWT_SECRET` masih bernilai default.

3.2 Docker Compose (Development)

Untuk menjalankan development environment lokal:

```
docker compose up -d      # Start PostgreSQL container
npm install                # Install dependencies
npx prisma db push        # Sync schema ke database
npx prisma db seed        # Seed data awal (SuperAdmin, AppSettings)
npm run dev                # Start Next.js dev server (port 4100)
```

3.3 Integrasi Google SSO

Konfigurasi Google OAuth Client ID:

1. Buka [Google Cloud Console](#) → [APIs & Services](#) → [Credentials](#)
2. Buat OAuth 2.0 Client ID (Application type: Web application)
3. Set Authorized JavaScript Origins: `http://localhost:4100` dan `https://fmsp-lintasarta.fly.dev`
4. Set Authorized Redirect URIs: `http://localhost:4100` dan `https://fmsp-lintasarta.fly.dev`
5. Salin Client ID ke `GOOGLE_CLIENT_ID` di `.env`

i Catatan: Google SSO memanfaatkan Google 2-Step Verification yang sudah aktif di akun Google Workspace pengguna, sehingga MFA otomatis terpenuhi.

3.4 Konfigurasi AI Copilot (Ollama)

- `OLLAMA_BASE_URL` : Alamat server Ollama (default: `http://127.0.0.1:11434`)
- `OLLAMA_MODEL` : Model yang digunakan (default: `qwen2.5:1.5b`)
- Status on/off dapat dikendalikan secara dinamis melalui AppSetting key `ai_bot_enabled`
- Fallback otomatis ke keyword search jika Ollama tidak merespons dalam 1.2 detik

4. Deployment & Infrastruktur

4.1 Fly.io Configuration (fly.toml)

Parameter	Nilai	Keterangan
app	fmsp-lintasarta	Nama aplikasi di Fly.io
primary_region	sin	Singapore — terdekat dengan Indonesia
internal_port	8080	Port internal container
force_https	true	Redirect HTTP → HTTPS
auto_stop_machines	stop	Hibernasi mesin saat idle
auto_start_machines	true	Bangun otomatis saat ada request
release_command	npx prisma@6.19.3 db push --accept-data-loss	Auto-migrate schema saat deploy
memory	1GB	RAM per VM instance
cpu_kind	shared	Shared CPU (cost-efficient)
health check path	/	GET / setiap 15 detik

4.2 Deployment Commands

```
# Deploy ke production
fly deploy

# Lihat status aplikasi
fly status

# Lihat logs real-time
fly logs
```

```
# SSH ke container
fly ssh console

# Set secrets (env vars di produksi)
fly secrets set JWT_SECRET="xxx" DATABASE_URL="xxx" CRON_SECRET="xxx"

# Lihat secrets yang sudah di-set
fly secrets list
```

4.3 Health Check & Auto-scaling

Fly.io melakukan health check HTTP GET ke path `/` setiap 15 detik dengan timeout 10 detik dan grace period 20 detik saat startup. Jika health check gagal, instance di-restart otomatis. Concurrency limit: soft=80, hard=100 koneksi per instance.

4.4 Fly.io Secrets Management

Semua variabel sensitif di-set sebagai Fly Secrets (bukan di fly.toml). Secrets yang wajib di-set:

```
fly secrets set \
  DATABASE_URL="postgresql://..." \
  JWT_SECRET="$(openssl rand -hex 64)" \
  CRON_SECRET="$(openssl rand -hex 32)" \
  GOOGLE_CLIENT_ID="xxx.apps.googleusercontent.com" \
  SMTP_HOST="smtp.office365.com" \
  SMTP_PORT="587" \
  SMTP_USER="noreply@lintasarta.co.id" \
  SMTP_PASSWORD="xxx" \
  NEXT_PUBLIC_VAPID_PUBLIC_KEY="xxx" \
  VAPID_PRIVATE_KEY="xxx" \
  VAPID_CONTACT="mailto:admin@lintasarta.co.id" \
  OLLAMA_BASE_URL="http://127.0.0.1:11434" \
  OLLAMA_MODEL="qwen2.5:1.5b"
```

5. Manajemen Database

5.1 Skema Database (22 Tabel)

Database PostgreSQL FMSP Lintasarta terdiri dari 22 model/tabel yang dikelola oleh Prisma ORM:

No	Model	Tabel PostgreSQL	Fungsi
1	User	users	Data pengguna, role, password hash, logout
2	PasswordResetRequest	password_reset_requests	Permintaan reset password (approval flow)
3	PushSubscription	push_subscriptions	Web Push subscription endpoints per user
4	Asset	assets	Aset fisik: gedung, tanah, kendaraan, fasilitas
5	AssetTransfer	asset_transfers	Riwayat mutasi lokasi aset
6	LegalDocument	legal_documents	Dokumen perizinan aset (9 tipe)
7	Notification	notifications	Email/dashboard notifications
8	MaintenanceSchedule	maintenance_schedules	Jadwal PM berkala per aset
9	VendorContract	vendor_contracts	Kontrak kerja sama vendor
10	Employee	employees	Data karyawan FM + Security (Gada/KTA)
11	InventoryItem	inventory_items	Stok gudang sparepart + safety stock
12	Smk3Item	smk3_items	Checklist item K3 + status inspeksi
13	AccountingTransaction	accounting_transactions	Jurnal transaksi keuangan

No	Model	Tabel PostgreSQL	Fungsi
14	RabBudget	rab_budgets	Pagu anggaran tahunan per departemen
15	WorkOrder	work_orders	Tiket kerusakan + approval workflow
16	HardwareDevice	hardware_devices	Perangkat IoT (BACnet/Modbus/MQTT)
17	DeviceState	device_states	State realtime perangkat IoT
18	TimeSeries	time_series	Data historis sensor (suhu, daya, PUE)
19	Alert	alerts	Alert severity dari sistem monitoring
20	AuditLog	audit_logs	Log audit operasi (user, action, IP)
21	MasterData	master_data	Dropdown dinamis (tipe aset, lokasi, dsb)
22	AppSetting	app_settings	Konfigurasi sistem (SMTP, company, bot)

5.2 Prisma Migrate & Client Generation

Prosedur saat ada perubahan schema:

```
# 1. Edit prisma/schema.prisma

# 2. Development - buat migration file
npx prisma migrate dev --name deskripsi_perubahan

# 3. Generate ulang Prisma Client
npx prisma generate

# 4. Production - sinkronisasi langsung
npx prisma db push --accept-data-loss
```

⚠ Perhatian: `--accept-data-loss` pada db push bisa menghapus data jika ada kolom yang dihapus. Selalu backup database sebelum migrasi di produksi!

5.3 Seeding Data Awal

Script `prisma/seed.ts` membuat data awal yang diperlukan:

- Akun SuperAdmin default (email dari `.env`)
- Master Data untuk semua dropdown kategori
- AppSettings default (SMTP config, company info, AI bot status)

```
npx prisma db seed
```

6. Prosedur Backup & Restore

6.1 Manual Backup (pg_dump)

Pencadangan database dilakukan dengan utilitas `pg_dump` :

Backup dari Database Lokal (Docker)

```
pg_dump -h localhost -U fmsp_user -d fmsp_db \  
-F c -b -v \  
-f backup/fmsp_backup_$(date +%Y%m%d_%H%M%S).dump
```

Backup dari Fly.io (Production)

```
# 1. Port-forward database Fly.io ke lokal  
fly proxy 5432 -a fmsp-lintasarta-db  
  
# 2. Jalankan pg_dump ke localhost  
pg_dump -h localhost -p 5432 -U fmsp_user -d fmsp_db \  
-F c -b -v \  
-f backup/fmsp_prod_$(date +%Y%m%d_%H%M%S).dump
```

6.2 Manual Restore (pg_restore)

```
pg_restore -v -h <host> -U <user> -d <db> \  
--clean --no-owner \  
/path/to/backup_file.dump
```

Flag `--clean` menghapus tabel lama sebelum restore. Flag `--no-owner` mencegah error ownership pada environment berbeda.

6.3 Backup Otomatis (Cron)

Direkomendasikan memasang cron job pada server database:

```
# Cron setiap hari pukul 01:00 WIB  
0 1 * * * /app/scripts/backup-db.sh >> /var/log/fmsp_backup.log 2>&1
```



```
# Retensi: simpan 30 backup terakhir, hapus yang lebih lama  
find /backup/ -name "fm*_*.dump" -mtime +30 -delete
```

7. Keamanan Sistem

7.1 Autentikasi & JWT

- Password di-hash menggunakan **bcrypt** dengan cost factor tinggi
- JWT ditandatangani dengan HS256 menggunakan `JWT_SECRET`
- Token berisi payload: `{ userId, email, role, region }`
- Expired time diatur pada saat penandatanganan
- Google SSO memverifikasi token via Google OAuth API endpoint

7.2 RBAC & Permission Matrix

Didefinisikan di `src/lib/rbac.ts` — 6 role dengan matriks permission per resource per action. Enforced oleh middleware `src/lib/rbac-middleware.ts` pada setiap API route.

7.3 Rate Limiting

Implementasi in-memory rate limiter per IP address:

- **Login endpoint:** 5 request per 1 menit
- **SSO endpoint:** 10 request per 1 menit
- **Upload endpoint:** 10 request per 1 menit
- **General API:** 60 request per 1 menit

7.4 XSS & Input Validation

Semua input API divalidasi oleh **Zod 4** schemas (`src/lib/validators.ts`). Fitur proteksi:

- String sanitization: strip tag `<script>` , escape HTML entities
- Email validation via regex ketat
- Enum validation untuk field status, role, tipe dokumen
- Number range validation (positif, max limit)
- File upload: MIME type whitelist, magic bytes verification, max 10MB

7.5 CORS & HTTP Headers

- CORS: Restricted ke domain yang terdaftar saja
- Error responses di-sanitize — tidak mengekspos stack trace atau path server
- API error handler terpusat (`src/lib/api-error.ts`) mengembalikan format JSON konsisten

8. API Endpoints Reference

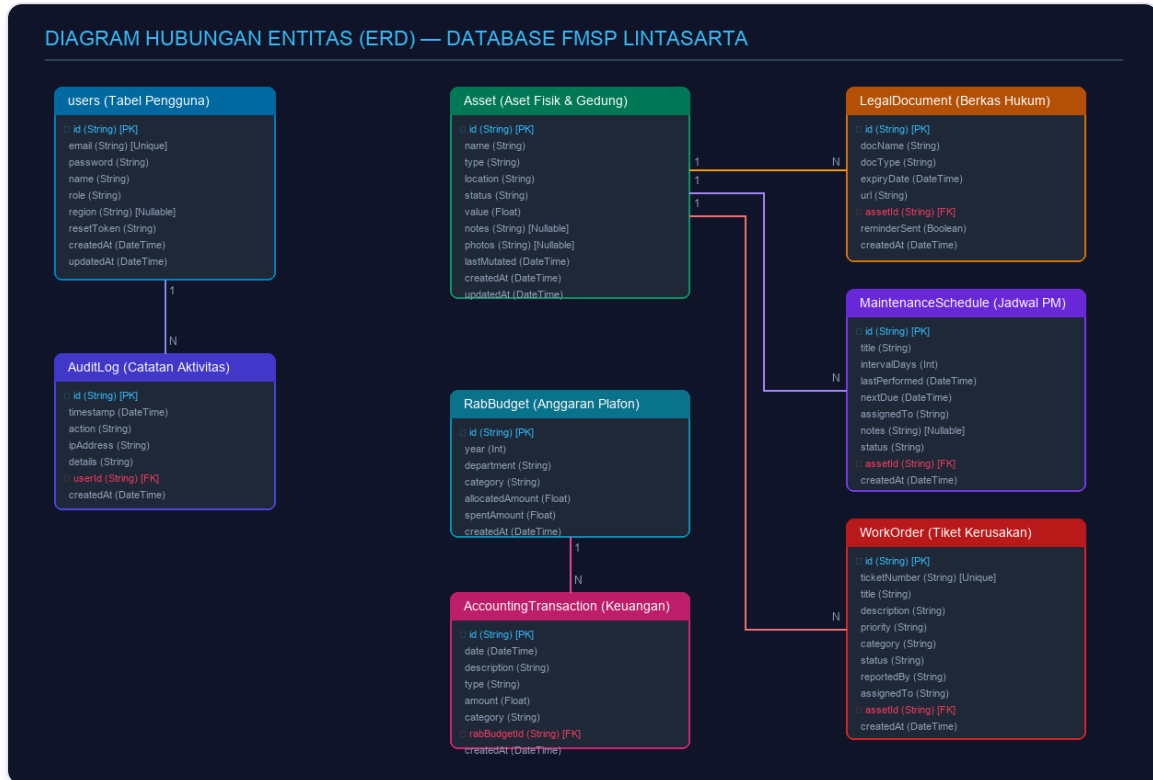
Method	Endpoint	Deskripsi	Auth
POST	/api/auth/login	Login email + password	Public
POST	/api/auth/sso	Login via Google SSO	Public
GET	/api/auth/me	Get current user profile	JWT
POST	/api/auth/forgot-password	Request password reset	Public
POST	/api/auth/reset-password	Execute password reset	Token
GET/POST/PUT/DELETE	/api/assets	CRUD aset fisik	JWT+RBAC
GET/POST/PUT/DELETE	/api/legal-documents	CRUD dokumen legalitas	JWT+RBAC
GET/POST	/api/notifications	Notifications CRUD	JWT
POST	/api/upload	File upload (10MB max)	JWT
GET/POST/PUT/DELETE	/api/management/maintenance	CRUD jadwal PM	JWT+RBAC
GET/POST/PUT/DELETE	/api/management/hrd	CRUD karyawan	JWT+RBAC

Method	Endpoint	Deskripsi	Auth
GET/POST/PUT/DELETE	/api/management/inventory	CRUD inventaris	JWT+RBAC
GET/POST/PUT	/api/management/smk3	CRUD checklist K3	JWT+RBAC
GET/POST	/api/management/accounting	CRUD transaksi keuangan	JWT+RBAC
GET/POST	/api/management/rab	CRUD pagu anggaran	JWT+RBAC
GET/POST/PUT	/api/management/workorder	CRUD work order + approval	JWT+RBAC
GET/POST/PUT	/api/management/vendor	CRUD vendor & kontrak	JWT+RBAC
GET/POST/PUT/DELETE	/api/management/users	Manajemen pengguna	JWT+Admin
GET/PUT	/api/management/admin	Master data CRUD	JWT+SuperAdmin
GET/PUT	/api/management/app-settings	App settings CRUD	JWT+SuperAdmin
GET/POST	/api/chat	AI Copilot (Ollama)	JWT
POST	/api/cron	Cron reminder processor	CRON_SECRET
GET	/api/analytics	Dashboard analytics data	JWT
GET	/api/audit-logs	Audit log viewer	JWT+Admin
GET	/api/reports	Compliance report	JWT

Method	Endpoint	Deskripsi	Auth
		generator	

9. Relasi Tabel Database (ERD)

9.1 Diagram ERD



Gambar 9.1: Diagram Hubungan Entitas (ERD) Database FMSP Lintasarta — 22 Tabel

9.2 Hubungan Foreign Key

Tabel Asal	Kolom FK	Tabel Referensi	ON DELETE	Kardinalitas
PasswordResetRequest	userId	User	CASCADE	N:1
PushSubscription	userId	User	CASCADE	N:1
AssetTransfer	assetId	Asset	CASCADE	N:1
LegalDocument	assetId	Asset	CASCADE	N:1
MaintenanceSchedule	assetId	Asset	CASCADE	N:1
WorkOrder	assetId	Asset	—	N:1 (optional)
AccountingTransaction	rabBudgetId	RabBudget	—	N:1 (optional)

9.3 Database Indexes

Index database dikonfigurasi pada kolom-kolom yang sering di-query untuk performa optimal:

Tabel	Index Columns	Tujuan
Asset	type, status, location	Filter aset per tipe/status/lokasi
LegalDocument	expiryDate, complianceStatus, assetId	Query dokumen yang akan expired
Notification	status, scheduledAt	Cron processor pending notifications
MaintenanceSchedule	nextDue, status	PM overdue checker
VendorContract	endDate, status	Kontrak expiring filter
WorkOrder	status, priority, assignedTo, approvalStatus	Work order queue management

Tabel	Index Columns	Tujuan
AuditLog	user, action, timestamp	Audit trail search
TimeSeries	deviceId + metric + timestamp	IoT data query optimization
Employee	role, department	Staff filter per jabatan
MasterData	category	Dropdown data lookup
AppSetting	group	Settings group query

10. Troubleshooting

Masalah	Penyebab	Solusi
Aplikasi tidak bisa start	JWT_SECRET masih default	Set JWT_SECRET unik: <code>openssl rand -hex 64</code>
Error "Cannot connect to database"	DATABASE_URL salah / PostgreSQL down	Verifikasi koneksi: <code>psql \$DATABASE_URL</code>
Google SSO error "Invalid client_id"	GOOGLE_CLIENT_ID tidak valid	Cek di Google Cloud Console → Credentials
Email notifikasi tidak terkirim	Konfigurasi SMTP salah	Test via Admin → Pengaturan SMTP → Kirim Test Email
User terkunci (locked)	5x gagal login	Tunggu 15 menit atau reset via Admin → Manajemen User
AI Copilot offline	Server Ollama tidak berjalan	<code>ollama serve</code> lalu <code>ollama pull qwen2.5:1.5b</code>
Migration error di Fly.io	Schema conflict	Backup → <code>fly ssh console</code> → <code>npx prisma db push --force-reset</code>
Upload gagal	File > 10MB atau tipe tidak diizinkan	Max 10MB. Tipe: PNG, JPG, PDF, DOCX
Cron tidak jalan	CRON_SECRET tidak diset	Set CRON_SECRET di .env dan di external cron caller
Push notification tidak muncul	VAPID keys tidak valid	Regenerate: <code>npx web-push generate-vapid-keys</code>

Kontak Tim Teknis

Tim	Kontak	Lingkup
IT Infrastructure	it-infra@lintasarta.co.id	Server, database, deployment, backup
DevOps	devops@lintasarta.co.id	CI/CD, Docker, Fly.io
Keamanan Informasi	infosec@lintasarta.co.id	Security audit, penetration testing